

Playstation 3 Consoles Tackle Black Hole Vibrations

When black holes are perturbed, they vibrate somewhat like a ringing bell. Now astronomers have narrowed down the rotational speed [at](#) which that vibration should stop.

As is typical, they did it out by running a simulation. But instead of a supercomputer, they used a batch of Sony Playstation 3 gaming consoles wired together.

The so-called PS3 Gravity Grid, a network of 16 Playstation 3 consoles grouped together in a cluster capable of running simulations that rival a dedicated supercomputer at a much lower cost.

?You can get a [supercomputer's capability](#) with relatively little money [at](#),? said Lior Burko of the University of Alabama, Huntsville, who led the black hole study, in an interview.

Rather than renting computer time on a supercomputer that could cost \$5,000 per simulation, Burko and his colleagues used the PS3 Gravity Grid built by Gaurav Khanna, a physics professor at the University of Massachusetts, Dartmouth.

The cluster can be built for around \$6,000, and allows simulations to be repeated at no extra cost. The system is tailored to simulations that require massive amounts of computations, but relatively little RAM memory.

Burko and Khanna used the PS3 Gravity Grid to run simulations that resolved an ongoing dispute over the speed at which spinning black holes stop vibrating [just after forming](#) or being perturbed by an outside object. One theory held that the black holes go silent at relatively fast speeds, while another theory contended they went quiet at slower speeds.

According to the new simulations, the gravity wave vibrations from a spinning black hole that's been perturbed would settle out according to the faster of the two theories,

though the actual speed will vary, Burko said.

"It depends on the mass of the black hole," he added. "Let's say you're in a spaceship orbiting the black hole and the black hole is perturbed. Then you need to wait a shorter time for the vibrations to settle to a certain amplitude."

The phenomena, Burko said, can be compared to a ringing bell.

"A bell rings, but eventually it gets quiet. The energy that goes out with the sound waves is energy that the bell is losing," Burko described in a statement. "A black hole does exactly that in gravitational waves instead of sound waves. A black hole that is wobbling is emitting gravitational waves. When those vibrations die down you get a quiet black hole."

Gravitational waves are predicted by Einstein's theory of general relativity to [emanate from neutron stars and black holes](#), but astronomers have not yet detected any directly.

The research is detailed in the Jan. 7 issue of the science journal *Classical and Quantum Gravity*.

While the PS3 Gravity Grid may not be useful for all types of research, for those requiring little RAM but massive computations it can prove a cost-cutting tool, said Khanna, who built a smaller cluster before stringing together the 16-machine grid.

"Science budgets  have been significantly dropping over the last decade," said Khanna, who describes how to build a PS3 computer  cluster on his Web site. "Here's a way that people can do science projects less expensively."

- Video - Black Hole Warping Time & Space
- Video - Black Hole Diving
- [Black Holes Preceded Galaxies, Discovery Suggests](#)